

Purchase Pattern Analysis

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Project Id : PTID-CDA-MAY-25-483A

Market basket Analysis

A data mining technique that is used to uncover purchase patterns in any retail setting is known as **Market Basket Analysis**. Basically, market basket analysis in data mining involves analyzing the combinations of products that are bought together.

This is a technique that gives the careful study of purchases done by a customer in a supermarket. This concept identifies the pattern of frequent purchase items by customers. This analysis can help to promote deals, offers, sale by the companies, and data mining techniques helps to achieve this analysis task. Example:

- Data mining concepts are in use for Sales and marketing to provide better customer service, to improve cross-selling opportunities, to increase direct mail response rates.
- Customer Retention in the form of pattern identification and prediction of likely defections is possible by Data mining.
- Risk Assessment and Fraud area also use the data-mining concept for identifying inappropriate or unusual behavior etc.

Market basket analysis mainly works with the ASSOCIATION RULE {IF} -> {THEN}.

- **IF** means **Antecedent**: An antecedent is an item found within the data
- **THEN** means **Consequent**: A consequent is an item found in combination with the antecedent.



Let's see ASSOCIATION RULE {IF} -> {THEN} rules used in Market Basket Analysis in Data Mining. For example, customers buying a domain means they definitely need extra plugins/extensions to make it easier for the users.

Like we said above Antecedent is the item sets that are available in data. By formulating from the rules means **{if}** component and from the example is the domain.

Same as Consequent is the item that is found with the combination of Antecedents. By formulating from the rules means **{THEN}** component and from the example is extra plugins/extensions.

With the help of these, we are able to predict customer behavioral patterns. From this, we are able to make certain combinations with offers that customers will probably buy those products. That will automatically increase the sales and revenue of the company.

APRIORI ALGORITHM

With the help of the [Apriori Algorithm](#), we can further classify and simplify the item sets which are frequently bought by the consumer.

There are three components in APRIORI ALGORITHM:

- SUPPORT
- CONFIDENCE
- LIFT

Example

Now take an example, suppose 5000 transactions have been made through a popular eCommerce website. Now they want to calculate the support, confidence, and lift for the two products, let's say pen and notebook for example out of 5000 transactions, 500 transactions for pen, 700 transactions for notebook, and 1000 transactions for both.

SUPPORT: It is been calculated with the number of transactions divided by the total number of transactions made,

$$\text{Support} = \text{freq}(A, B) / N$$

support(pen) = transactions related to pen / total transactions

i.e support $\rightarrow 500 / 5000 = 10$ percent

CONFIDENCE: It is been calculated for whether the product sales are popular on individual sales or through combined sales. That is calculated with combined transactions / individual transactions.

$$\text{Confidence} = \text{freq}(A, B) / \text{freq}(A)$$

Confidence = combine transactions / individual transactions

i.e confidence $\rightarrow 1000 / 500 = 20$ percent

LIFT: Lift is calculated for knowing the ratio for the sales.

$$\text{Lift} = \text{confidence percent} / \text{support percent}$$

Lift $\rightarrow 20 / 10 = 2$

When the Lift value is below 1 means the combination is not so frequently bought by consumers. But in this case, it shows that the probability of buying both the things together is high when compared to the transaction for the individual items sold.

With this, we come to an overall view of the Market Basket Analysis in Data Mining and how to calculate the sales for combination products.

Types of Market Basket Analysis

There are three types of Market Basket Analysis. They are as follow:

1. **Descriptive market basket analysis:** This sort of analysis looks for patterns and connections in the data that exist between the components of a market basket. This kind of study is mostly used to understand consumer behavior, including what products are purchased in combination and what the most typical item combinations. Retailers can place products in their stores more profitably by understanding which products are frequently bought together with the aid of descriptive market basket analysis.
2. **Predictive Market Basket Analysis:** Market basket analysis that predicts future purchases based on past purchasing patterns is known as predictive market basket analysis. Large volumes of data are analyzed using machine learning algorithms in this sort of analysis in order to create predictions about which products are most likely to be bought together in the future. Retailers may make data-driven decisions about which products to carry, how to price them, and how to optimize shop layouts with the use of predictive market basket research.
3. **Differential Market Basket Analysis:** Differential market basket analysis analyses two sets of market basket data to identify variations between them. Comparing the behavior of various client segments or the behavior of customers over time is a common usage for this kind of study. Retailers can respond to shifting consumer behavior by modifying their marketing and sales tactics with the help of differential market basket analysis.

Benefits of Market Basket Analysis

- 1. Enhanced Customer Understanding:** Market basket research offers insights into customer behavior, including what products they buy together and which products they buy the most frequently. Retailers can use this information to better understand their customers and make informed decisions.
- 2. Improved Inventory Management:** By examining market basket data, retailers can determine which products are sluggish sellers and which ones are commonly bought together. Retailers can use this information to make well-informed choices about what products to stock and how to manage their inventory most effectively.
- 3. Better Pricing Strategies:** A better understanding of the connection between product prices and consumer behavior might help merchants develop better pricing strategies. Using this knowledge, pricing plans that boost sales and profitability can be created.
- 4. Sales Growth:** Market basket analysis can assist businesses in determining which products are most frequently bought together and where they should be positioned in the store to grow sales. Retailers may boost revenue and enhance customer shopping experiences by improving store layouts and product positioning.

Dataset (cleaned once)

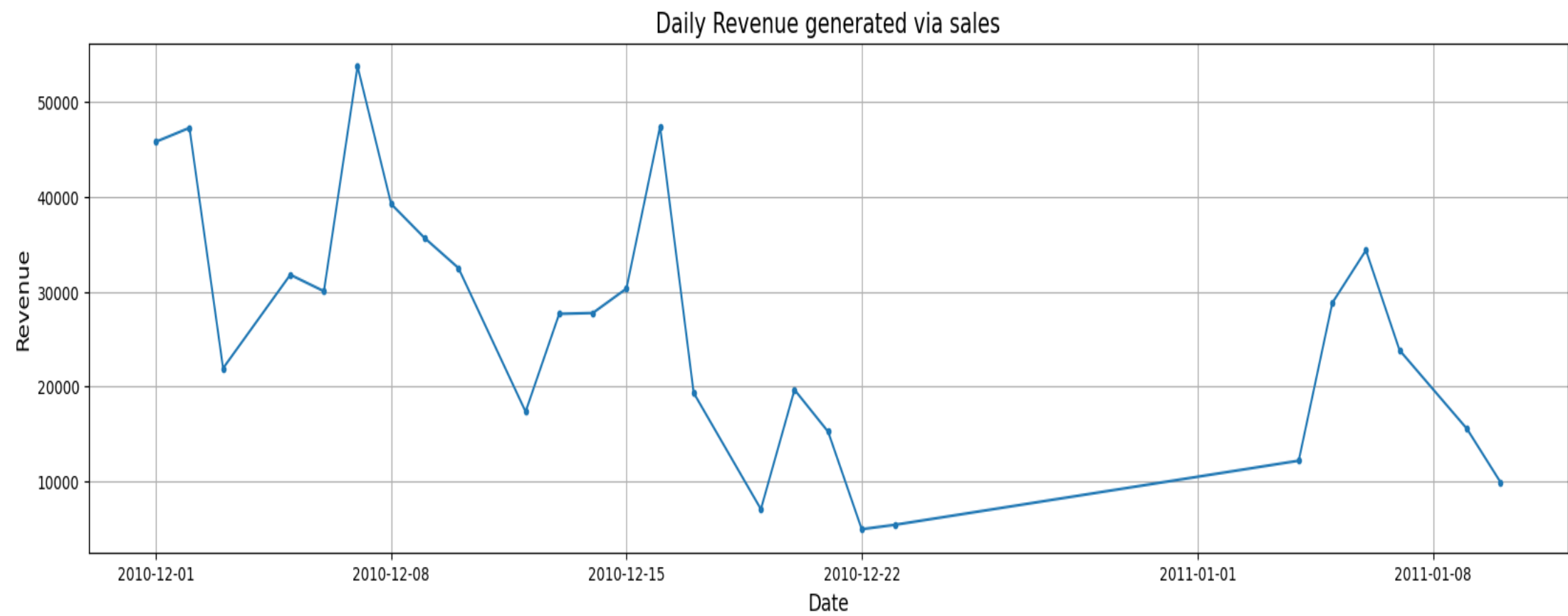
	BillNo	Itemname	Quantity	Present_Date	Price	CustomerID	Country	Revenue
0	536365	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850	United Kingdom	15.30
1	536365	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850	United Kingdom	20.34
2	536365	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850	United Kingdom	22.00
3	536365	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850	United Kingdom	20.34
4	536365	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850	United Kingdom	20.34
...	...	...	...	...	...	...	...	...
31427	540645	HOME BUILDING BLOCK WORD	18	2011-01-10 14:20:00	5.45	14060	United Kingdom	98.10
31428	540645	TEA PARTY BIRTHDAY CARD	24	2011-01-10 14:20:00	0.42	14060	United Kingdom	10.08
31429	540645	FANCY FONT BIRTHDAY CARD,	24	2011-01-10 14:20:00	0.42	14060	United Kingdom	10.08
31430	540645	CARD DOG AND BALL	24	2011-01-10 14:20:00	0.42	14060	United Kingdom	10.08
31431	540645	CARD CAT AND TREE	24	2011-01-10 14:20:00	0.42	14060	United Kingdom	10.08

31432 rows × 8 columns

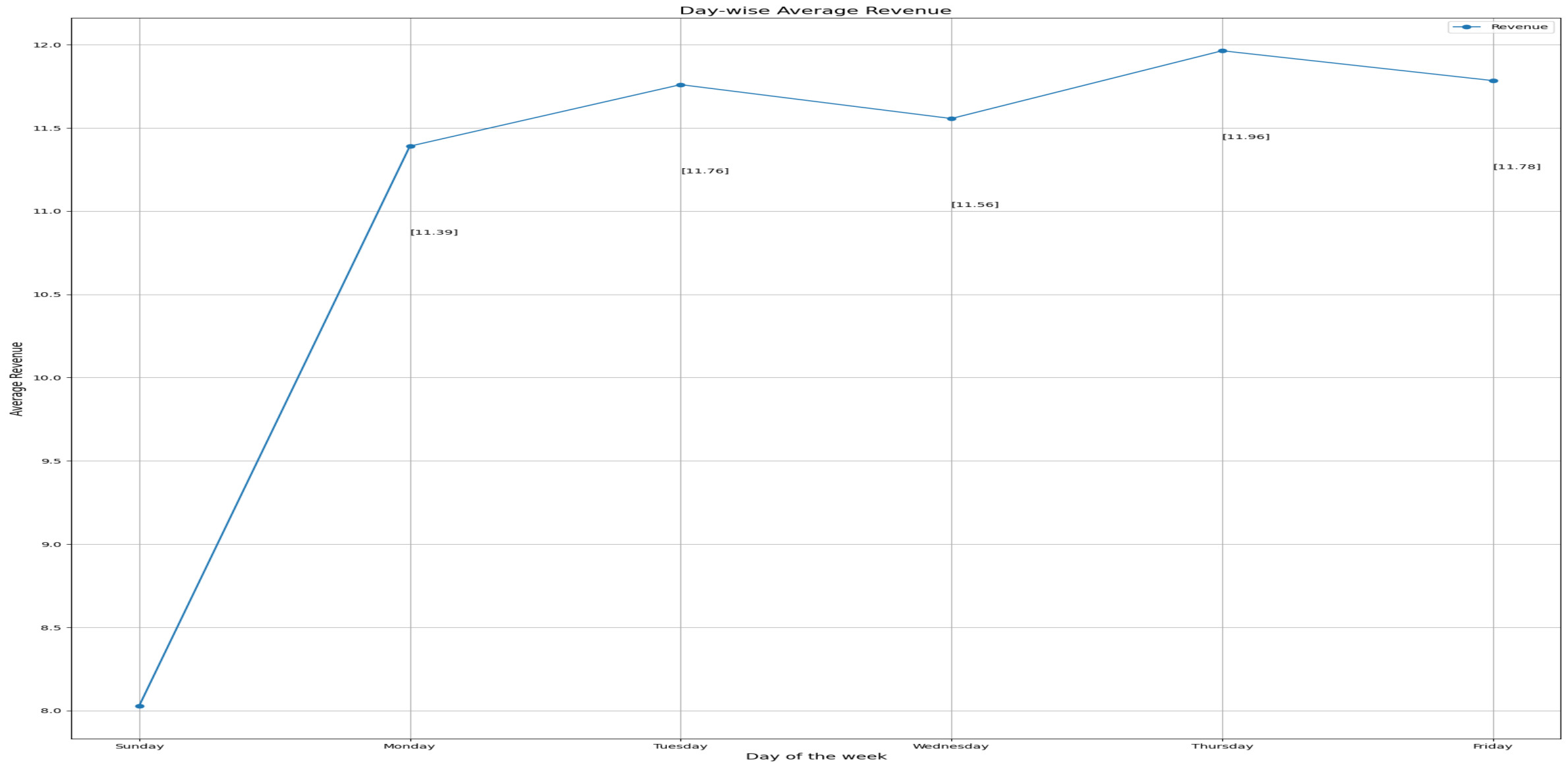


Data visualization

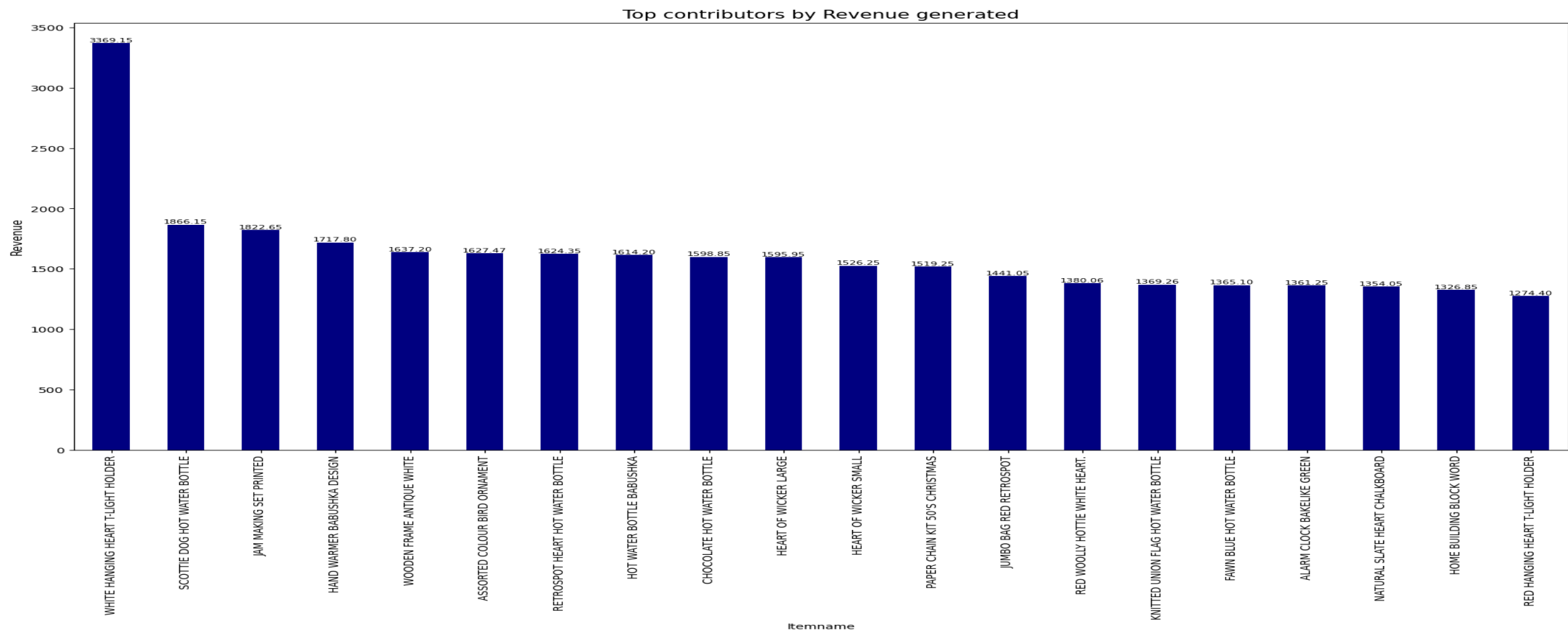
Daily Sales Distribution :



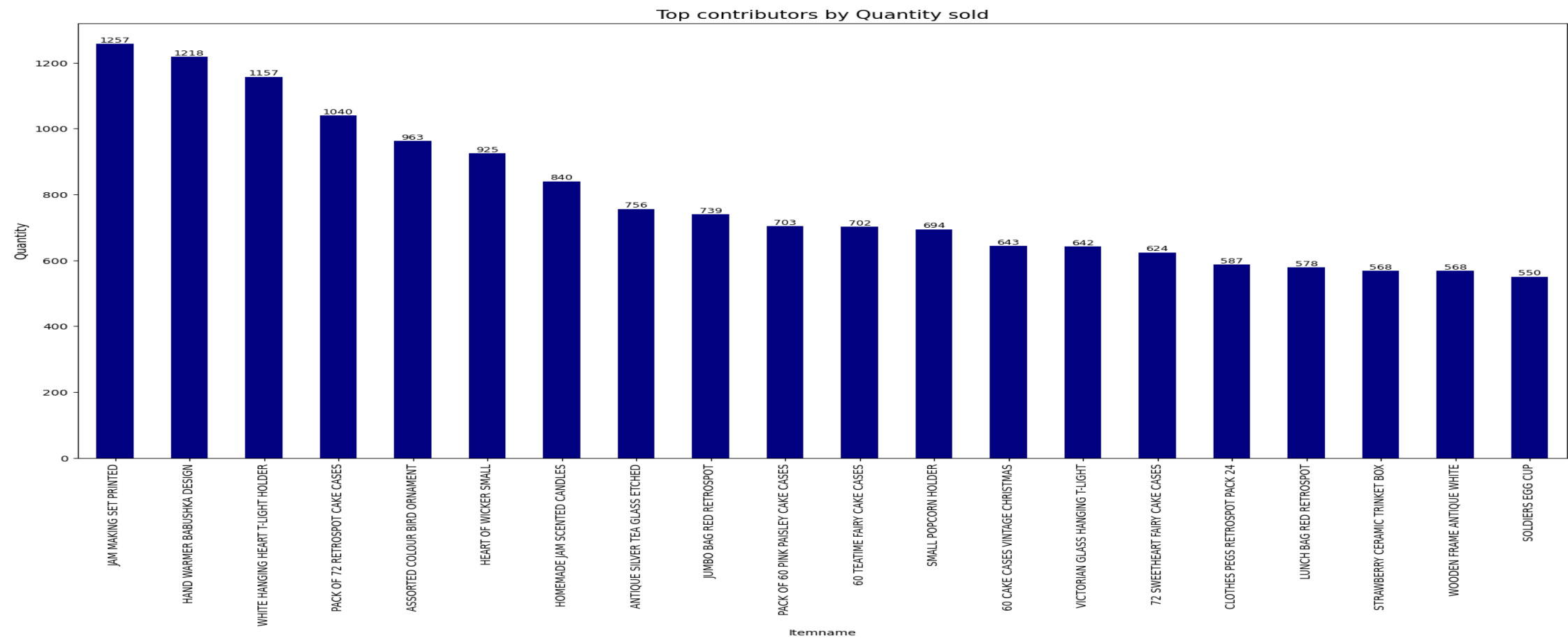
Distribution of Revenue as per day-wise :



Distribution of Itemnames as per Revenue :



Distribution of Itemnames as per Quantity :



Dashboard





Association Rule Mining (apriori algorithm)

```
[14] # Association Rule Mining
frequent_itemsets = apriori(basket_encoded, min_support=0.01, use_colnames=True)
rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=0.6)
```

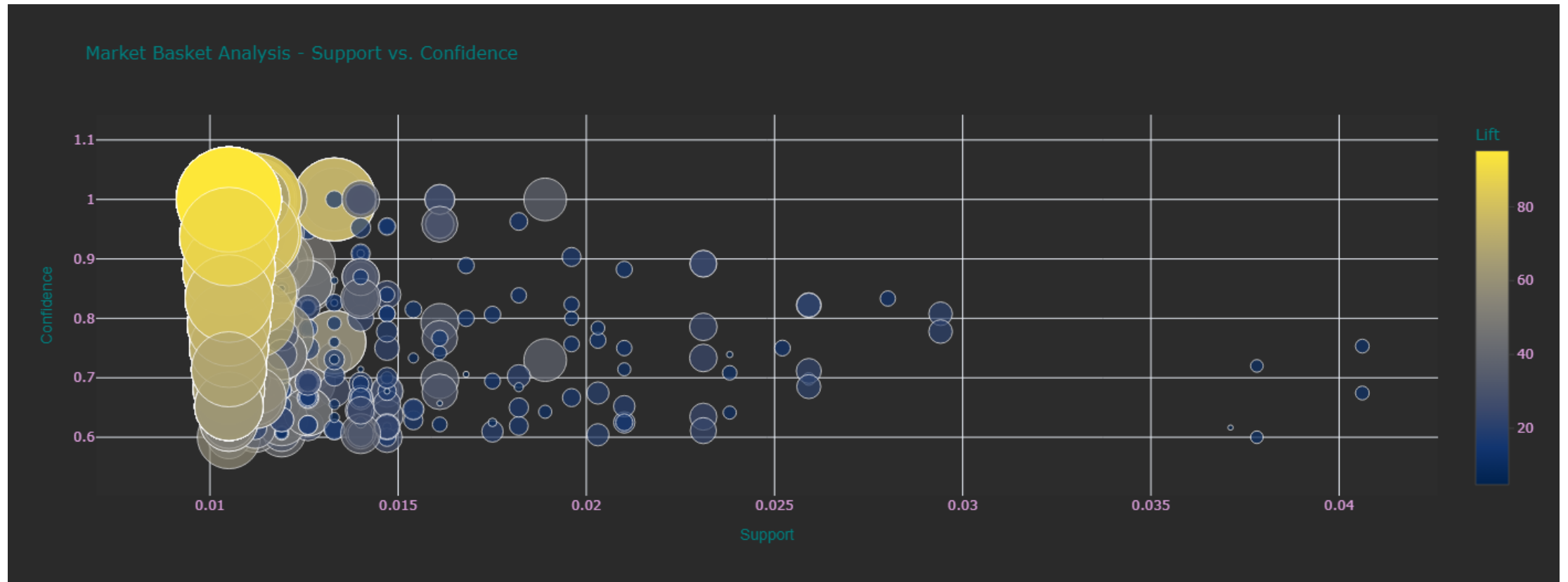
```
[15] # Display information of the rules
print("Association Rules:")
rules.head()
```



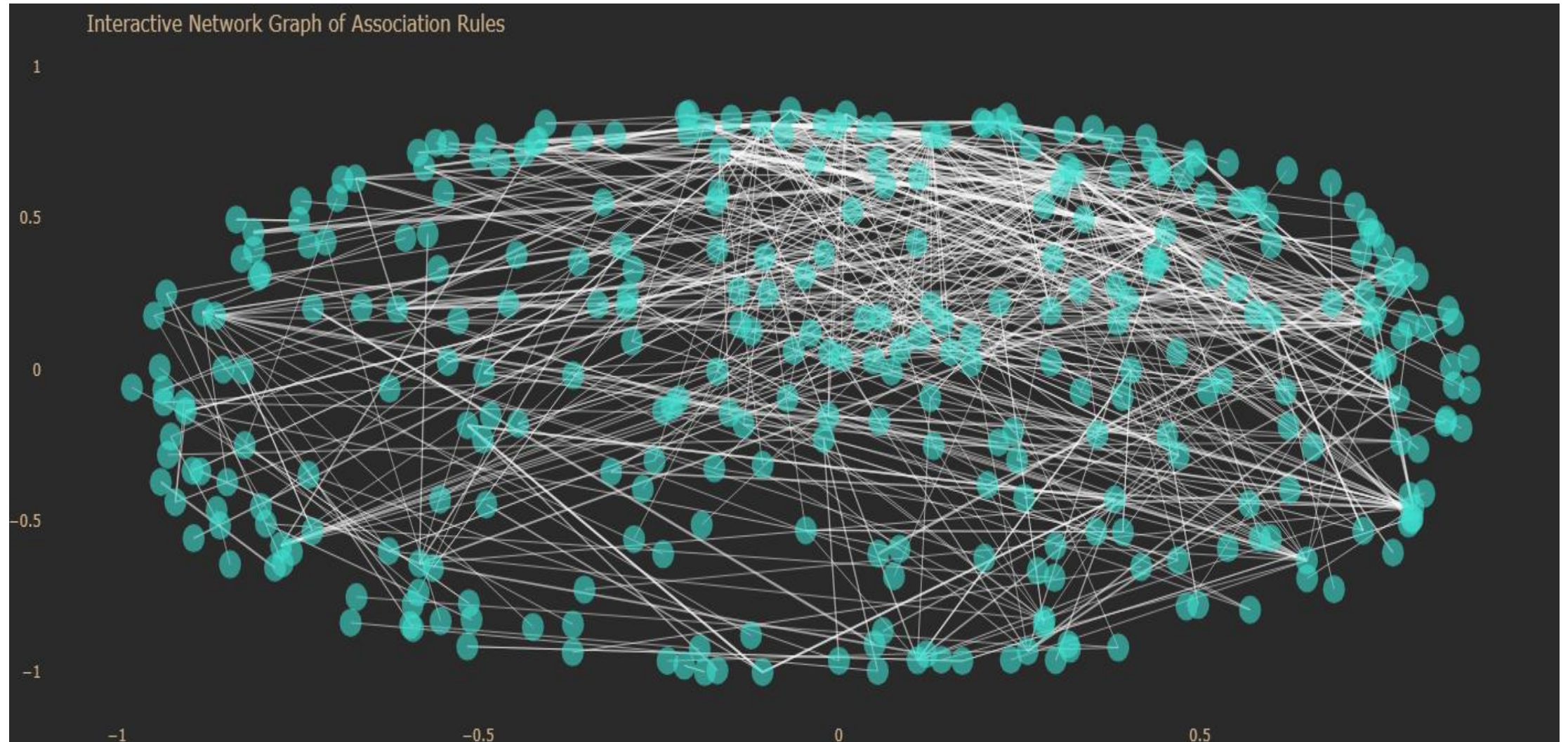
Association Rules:

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	representativity	leverage	conviction	zhangs_metric	jaccard	certainty	kulczynski
0	(12 PENCILS SMALL TUBE RED RETROSPOT)	(12 PENCILS SMALL TUBE SKULL)	0.023941	0.021486	0.014733	0.615385	28.641758	1.0	0.014219	2.544138	0.988758	0.480000	0.606939	0.650549
1	(12 PENCILS SMALL TUBE SKULL)	(12 PENCILS SMALL TUBE RED RETROSPOT)	0.021486	0.023941	0.014733	0.685714	28.641758	1.0	0.014219	3.105642	0.986277	0.480000	0.678005	0.650549
2	(ALARM CLOCK BAKELIKE CHOCOLATE)	(ALARM CLOCK BAKELIKE GREEN)	0.019644	0.058318	0.016575	0.843750	14.468092	1.0	0.015429	6.026765	0.949535	0.270000	0.834074	0.563980
3	(ALARM CLOCK BAKELIKE CHOCOLATE)	(ALARM CLOCK BAKELIKE RED)	0.019644	0.053407	0.016575	0.843750	15.798491	1.0	0.015525	6.058195	0.955472	0.293478	0.834934	0.577047
4	(ALARM CLOCK BAKELIKE IVORY)	(ALARM CLOCK BAKELIKE GREEN)	0.028238	0.058318	0.020258	0.717391	12.301373	1.0	0.018611	3.332106	0.945405	0.305556	0.699889	0.532380

Market Basket Analysis – Support vs. Confidence



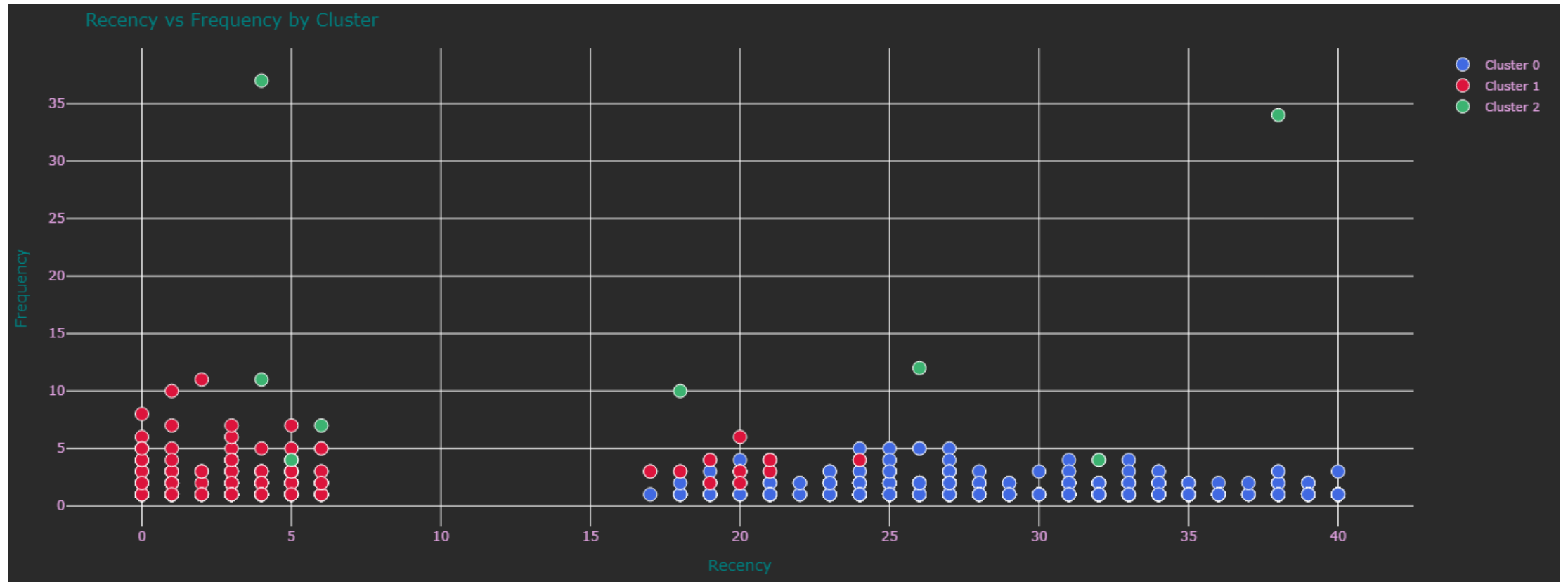
Interactive Network Graph of Association Rules



RFM Analysis

	CustomerID	recency	frequency	monetary
0	12347	33	1	711.79
1	12370	24	1	277.20
2	12377	21	1	1001.52
3	12383	18	1	600.72
4	12386	4	2	401.90

Recency vs Frequency by Cluster



Insights

- Sum of Sales trended down, resulting in a 85.32% decrease between Friday, January 1, 2010 and Saturday, January 23, 2010.
- Sum of Sales started trending down on Wednesday, January 13, 2010, falling by 78.93% (20070.44) in 10 days.
- Sum of Sales dropped from 25429.56 to 5359.12 during its steepest decline between Wednesday, January 13, 2010 and Saturday, January 23, 2010.
- Sum of Price was highest for WHITE HANGING HEART T-LIGHT HOLDER at 713.25, followed by PAPER CHAIN KIT 50'S CHRISTMAS and HEART OF WICKER LARGE.
- WHITE HANGING HEART T-LIGHT HOLDER accounted for 2.07% of Sum of Price.
- Across all 1,801 Itemname, Sum of Price ranged from 0.00e+0 to 713.25.
- WHITE HANGING HEART T-LIGHT HOLDER had the highest Sum of Sales (7448.25) and HAND WARMER BABUSHKA DESIGN had the highest Sum of Quantity (3030).

Recommendations

- Product Bundling : Bundle “T-light holder + chalkboard” or “Hand warmer sets” combo discount.
- Cross-Sell at Checkout : Suggest “ lunch bag skull” with red version display “building block word” with “party bunting”.
- Curated Gift Sets : Create seasonal kits (cozy winter, kitchen starter); highlight collections on homepage.
- Personalized Recommendations : Build A recommender using association rules; show “Others bought...” widgets.